Question 1: What is Reinforcement Learning (RL)? Discuss.

Reinforcement learning is a type of machine learning where a self-governing agent learns through trial and error, using numerical values to reward or penalize its actions to determine the optimal decisions over time. This type of learning works in a continuous loop between the agent and its environment, beginning with data which provides information about the current state of the environment. From there, the agent uses that information to choose an action (decision), which then transitions into a new state where it receives a reward or penalty based on the action taken. This process continues as the agent learns from its previous actions and the feedback it receives. Essentially, it creates a balance between trying and learning unknown behaviors and using previously learned or known behaviors to make the most optimal decisions based on what it knows.

Question 2: What are the real-world applications which use RL? Discuss two application areas.

Two very straightforward real-world applications that use reinforcement learning are healthcare and robotics.

Healthcare: In the healthcare field, researchers use reinforcement learning to study how AI can help determine things like medication dosages and treatment decisions. A specific example of this would be diabetes treatment, where the agent could recommend insulin doses based on the patient's changing health conditions. The environment would involve the patient's response to treatment, while the state would include patient information, such as blood glucose levels. In this case, a reward would be the patient's blood glucose remaining within a safe medical standard, while a penalty would be levels falling outside that healthy standard. Through this process, the agent learns which treatment decisions produce better outcomes over time. These recommendations would still need to be evaluated and monitored by healthcare professionals to ensure patient safety.

Robotics: Reinforcement learning can also be used to train robots to perform certain tasks through trial and error. For example, if we wanted to train a robot to approach a table and pick up a glass of water, the agent would be the robot being trained, and the environment would be its surroundings. The state would include the robot's current position, movement, and the location of the glass. The actions would involve how the robot approaches the table, which route it takes, and how it uses its arms or hands to pick up the glass. A reward would be successfully picking up the glass and bringing it back, while penalties could include knocking over the glass, failing to pick it up, or taking an inefficient route. Things like the robot losing balance or tipping over could also be part of the trial-and-error process. Through repeated training, the robot can learn which movements produce successful outcomes and, over time, improve its ability to complete these tasks independently.

Question 3: Discuss the advantages and disadvantages of Reinforcement Learning.

When it comes to reinforcement learning, we think that sometimes the advantages can outweigh the disadvantages, but we also believe that this is case-specific and depends on the situation in which it is being applied.

One advantage of reinforcement learning is that it learns through experience. Reinforcement learning allows an agent to continuously learn through trial and error rather than being limited to one definite answer or a set of predefined decisions. This allows for more possibilities and adaptability. Another advantage is improved decision-making over time. Since reinforcement learning works in a continuous loop, the agent can use what it has learned from previous actions to make better decisions and produce better outcomes. We also think another advantage is its ability to solve complex problems, especially those involving multiple factors that need to be considered. For example, in this project's navigation scenario, factors such as toll roads, highways, local routes, and travel time could all influence which route is considered the most optimal.

As for disadvantages, we would say that reinforcement learning can be very time-consuming. An agent may require a large amount of training experience and numerous attempts, whether using the same environment or different situations, before it learns an effective strategy. This can also make it less cost-effective because training may require significant computing power and resources, including electricity and, depending on the computing infrastructure, as we all know, water for cooling. Another disadvantage is the possibility of mistakes during trial and error, especially when applied to real-world situations such as healthcare. This is why we think having a second step of human evaluation is important, because the agent's decisions are limited by what it has learned and the information available to it.

A good example of both an advantage and disadvantage would be the robotics application we previously discussed. An advantage is that the robot can gradually learn the best way to perform a task, such as picking up a glass of water. It can learn which route, movements, and positioning of its arms or hands work best for successfully grabbing the glass. However, a disadvantage is that the robot might knock over the glass 50 or even 1,000 times before successfully picking it up. Even after succeeding, that does not guarantee that it will successfully complete the task every time afterward. Essentially, this goes back to our original point that the advantages and disadvantages of reinforcement learning can be case-specific.